Xiangpeng Yang

Ph.D. Researcher in Foundation Models, Generative AI and Multi-modal Learning

Email: Xiangpeng.Yang@student.uts.edu.au | Homepage | *WeChat:* 13915969870 | Google Scholar | GitHub

Research Profile

Incoming Student Researcher at **Google** and Ph.D. student building **generative image/video foundation models**. At Alibaba, I independently developed **Infinity Shots**, the first 60-second multi-shot audio-video model, and helped **Z-Image 1.5** surpass Seedream 5.0. I developed **VideoCoF** (CVPR 2026 **Highlight**, Top 0.65%) and **VideoGrain** (ICLR 2025) among 10+ top-tier papers.

Research Internship

Google *Fall 2026*

Incoming Student Researcher

- Working with Dr. Yale Song on video editing for Gemini Omni (previously Veo 3.1).

Tongyi MAI Lab, Alibaba *Mar 2026 - Present*

*Foundation Model Research Intern, **Z-Image Team*** *Hangzhou, China*

- Supervised by Dr. Peng Gao, Dr. Zhen Li, and Prof. Steven Hoi; core contributor to **Infinity Shots**, **Z-Image 1.5**, **D-OPSD**, and **Z-Reward**.
- **Sole contributor to Infinity Shots (Z-Multi-Shot Video)**, the first 60-second multi-shot audio-video generation model; built the end-to-end data curation, multi-shot stitching, and context-parallel training pipelines from scratch over 14M one-minute videos.
- Core contributor to **Z-Image 1.5**: building prompt enhancement, raising the human GSB win rate from 42.0% → 58.5%, surpassing Seedream 5.0; developed an automated multi-subject identity-consistency pipeline for Z-Image Edit by extracting image pairs from multi-shot videos.

Baidu VIS *Aug 2022 - Mar 2023*

Research Intern *Beijing, China*

- Proposed a dynamic global-local (DGL) prompt tuning for T2V retrieval, surpassing full finetuning/PEFL methods with tuning only 0.83 MB parameters. Accepted at AAAI 2024. [paper] [video]

Representative Works

[1]. Infinity Shots: Chunk-wise Learning for Infinite Multi-Shot Audio-Video Generation
Project Page
Xiangpeng Yang
Technical Report, 2026
TL;DR: First 60s multi-shot audio-video model (2× Seedance 2.5/Wan 3.0), beating open-source SOTA JoyAI-Echo; chunk-wise learning for second-level control of 2–30 shots in 1min over 14 genres.

[2]. VideoCoF: Unified Video Editing with Temporal Reasoner
arXiv Project Page Code Hugging Face Video
***Xiangpeng Yang**, Ji Xie, Yiyuan Yang, Yue Ma, Min Xu, Qiang Wu*
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026, Highlight, Top 0.65%
Impact: The first Chain-of-Frames framework for unified video editing, validating that video generative models can reason over video content and achieve 16× length extrapolation to 512+ frames.

[3]. VideoGrain: Modulating Space-Time Attention for Multi-Grained Video Editing
arXiv Project Page Code Video

***Xiangpeng Yang**, Linchao Zhu, Hehe Fan, Yi Yang*
International Conference on Learning Representations (ICLR), 2025
Highlight: The first work to introduce multi-grained video editing at the class, instance, and part levels through space-time attention modulation.

[4]. **DGL: Dynamic Global-Local Prompt Tuning for Text-Video Retrieval**
arXiv Code Video
***Xiangpeng Yang**, Linchao Zhu, Xiaohan Wang, Yi Yang*
AAAI Conference on Artificial Intelligence (AAAI), 2024
Highlight: Outperforms full fine-tuning while updating only 0.83 MB of parameters.

[5]. **EffiVMT: Video Motion Transfer via Efficient Spatial-Temporal Decoupled Finetuning**
arXiv Project Page *Equal contribution; §Project lead.
Yue Ma, Yulong Liu*, **Xiangpeng Yang***§, Qifeng Chen*
International Conference on Learning Representations (ICLR), 2026

[6]. **D-OPSD: On-Policy Self-Distillation for Continuously Tuning Step-Distilled Diffusion Models**
arXiv Project Page Code
*Dengyang Jiang, …, **Xiangpeng Yang**, Peng Gao, Steven C. H. Hoi*
Advances in Neural Information Processing Systems (NeurIPS), 2026
Highlight: The first on-policy distillation method for few-step diffusion models, enabling continuous learning of new concepts, styles, and domain preferences while preserving few-step knowledge.

[7]. **ActAlign: Action-Aware Representation Alignment for Generalizable Vision-Language-Action Models**
Paper Demo
*Yiyuan Yang, Guodong Long, **Xiangpeng Yang**, Dikai Liu, Qinghua Lu, Jing Jiang*
AAAI Conference on Artificial Intelligence (AAAI), 2027 In Submission
Highlight: Proposed action-aware representation alignment for embodied VLA models.

[8]. **Z-Reward: Beyond Scalar Rewards by Internalizing Reasoning into Score Distributions**
arXiv
*Xin Jin, Huanqia Cai, …, **Xiangpeng Yang**, …, Steven C. H. Hoi*
Z-Image Technical Report, 2026

[9]. **An Empirical Study of Training Pixel-Space Text-to-Image Diffusion Models**
arXiv
*Dengyang Jiang, …, **Xiangpeng Yang**, Peng Gao, Harry Yang, Steven C. H. Hoi*
AAAI Conference on Artificial Intelligence (AAAI), 2027 In Submission
Highlight: A practical latent-to-pixel training recipe that matches or outperforms latent-space counterparts while achieving 3.18–4.75× end-to-end inference speedups.

[10]. **Group Editing: Edit Multiple Images in One Go**
arXiv Project Page Code
Yue Ma, Xinyu Wang*, Qianli Ma, **Xiangpeng Yang**, Qifeng Chen*
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026

[11]. **EasyCreator: Empowering 4D Creation through Video Inpainting**
arXiv Project Page
Yue Ma, Kunyu Feng*, …, **Xiangpeng Yang**, Qifeng Chen*
International Conference on Learning Representations (ICLR), 2026

Education

University of Technology Sydney (UTS) *Apr 2023 - present*
Advisers: Prof. Qiang Wu, Prof. Yi Yang, and Prof. Linchao Zhu
Ph.D. in Computer Science *Sydney, Australia*

Southeast University (SEU) *Sep 2019 - Jun 2022*
M.S. in Automation *Nanjing, China*

Northeastern University (NEU) *Sep 2015 - Jun 2019*
B.S. in Automation (Honors Class) *Shenyang, China*

Technical Expertise

Modeling	Video Foundation Models, Diffusion Transformers (DiT), Multimodal Generation, Multi-shot Video Generation, Unified Video Editing
Systems & Data	Foundation-Scale Video Data Curation, Distributed VDM Training on 100+ H100 GPUs, FSDP, Context Parallelism, Training Infrastructure
Post-Training	Supervised Fine-Tuning, Few-Step Distillation, Reinforcement Learning, On-Policy Distillation, Reward-Guided Training
Applications	Multi-Grained Video Editing, Motion Transfer, 4D Creation, Text-to-Video Retrieval

Selective Awards

- 2023-2027 UTS ARC Discovery Scholarship & International Research Scholarship
- 2020 Baidu AI Studio-Lane Recognition Algorithm Competition, *Top 3/1020*

Academic service

Regular Reviewer for Conferences CVPR, ICLR, ICML, NeurIPS, ICCV, ECCV